

Imerslund-Gräsbeck Syndrome Type 1

(IGS-1): Comprehensive Disease Characteristics Report

Disease: Imerslund-Gräsbeck Syndrome Type 1 (Megaloblastic Anemia 1, MGA1) **MONDO ID:** MONDO:0100156 · **OMIM:** #261100 · **Category:** Mendelian (autosomal recessive) **Causal gene:** *CUBN* (cubilin), chromosome 10p12.1

Summary

Imerslund-Gräsbeck Syndrome type 1 (IGS-1; also called megaloblastic anemia 1, MGA1) is a rare autosomal recessive disorder of **selective intestinal vitamin B12 (cobalamin) malabsorption** caused by biallelic pathogenic variants in *CUBN*, the gene encoding **cubilin**. Cubilin is a 460-kDa multiligand peripheral membrane endocytic receptor that, together with its obligate partner **amnionless (AMN)**, forms the **cubam** receptor complex. Cubam mediates two physiologically distinct functions: (1) uptake of the **intrinsic factor–vitamin B12 (IF–Cbl)** complex across the ileal brush border, and (2) reabsorption of filtered low-molecular-weight plasma proteins (albumin, transferrin, vitamin D-binding protein, apolipoprotein A-I) in the **renal proximal tubule**. Loss of cubilin function therefore produces the syndrome's two hallmark features: **selective B12 malabsorption not correctable by exogenous intrinsic factor**, and **benign low-molecular-weight proteinuria**.

Clinically, IGS-1 presents in infancy or early childhood (typically from ~4 months up to several years of age) with **megaloblastic anemia, failure to thrive, recurrent infections**, and—if untreated—**neurological/neurocognitive damage** that can be the sole presenting manifestation. Mild proteinuria without renal insufficiency is present in roughly half of patients. The metabolic signature of intracellular B12 deficiency (methylmalonic aciduria and hyperhomocysteinemia) accompanies the hematologic picture. Prevalence is approximately **1:200,000**, with the highest rates in Finland (0.8/100,000) and Norway due to founder effects, and additional clusters in Middle Eastern countries driven by consanguinity. IGS type 2 is the allelic disorder caused by *AMN* mutations (chromosome 14).

The disorder is exquisitely treatable: **lifelong parenteral (intramuscular) hydroxocobalamin/cyanocobalamin** fully corrects the anemia and metabolic derangement, and early treatment allows complete recovery. Maintenance dosing as infrequent as 1 mg twice yearly can suffice. Prognosis is excellent with timely diagnosis, but **delayed diagnosis can leave permanent neurodevelopmental deficits**, and the proteinuria persists lifelong (though benign). The molecular defect is well characterized at atomic resolution: the Finnish founder mutation **p.Pro1297Leu (FM1)** lies in the IF–Cbl-binding region (CUB domains 5–8) and impairs IF–Cbl recognition, and the crystal structure of IF–Cbl bound to cubilin CUB5–8 reveals a Ca^{2+} -dependent "dual-point" binding mode. Naturally occurring canine IGS models (Beagles, Border Collies) faithfully recapitulate the human disease.

Key Findings

F001 — IGS-1 is an autosomal recessive selective B12 malabsorption disorder caused by biallelic *CUBN* mutations

IGS type 1 (megaloblastic anemia 1, MGA1; OMIM #261100; MONDO:0100156) is caused by biallelic loss-of-function and missense mutations in **CUBN** on chromosome **10p12.1**, encoding cubilin, the intrinsic factor–vitamin B12 receptor. The landmark linkage and positional cloning study by Aminoff et al. (1999) refined the MGA1 locus by linkage-disequilibrium mapping, fine-mapped *CUBN*, and identified two independent disease-specific *CUBN* mutations across 17 Finnish MGA1 families, establishing *CUBN* as causal:

"We have now refined the MGA1 region by linkage disequilibrium (LD) mapping, fine-mapped CUBN and identified two independent disease-specific CUBN mutations in 17 Finnish MGA1 families. Our genetic and molecular data indicate that mutations in CUBN cause MGA1." — [PMID: 10080186](#)

Genetic heterogeneity underlies the syndrome: IGS type 1 results from *CUBN* variants (chr 10), whereas the clinically indistinguishable IGS type 2 results from *AMN* variants (chr 14):

"the molecular basis of the selective malabsorption and proteinuria involves a mutation in one of two genes, cubilin (CUBN) on chromosome 10 or amnionless (AMN) on chromosome 14" — [PMID: 16722557](#)

Ontology/annotation suggestions: MONDO:0100156; HGNC gene *CUBN*; inheritance HP:0000007 (autosomal recessive).

F002 — Cubam (cubilin + amnionless) is a dual-tissue endocytic receptor: ileal B12 uptake and renal protein reabsorption

Cubilin is a large peripheral membrane endocytic receptor containing **27 CUB domains** for ligand binding. It has no transmembrane anchor of its own and depends on the membrane protein **amnionless (AMN)** for correct apical membrane translocation—together they constitute the **cubam** complex. In the renal proximal tubule, cubilin additionally cooperates with **megalin (LRP2)**. Its ligand repertoire includes IF-B12, albumin, transferrin, and vitamin D-binding protein:

"Cubilin is a large endocytic receptor serving such diverse functions as the intestinal absorption of the intrinsic factor-B(12) complex and the renal proximal tubule reabsorption of filtered proteins including albumin, transferrin, vitamin D-binding protein and other important plasma carriers. Cubilin is a structurally unique, peripheral membrane protein, which depends on the membrane protein amnionless (AMN) for correct apical translocation." — [PMID: 23291372](#)

"CUBN encodes for cubilin, an intestinal and proximal tubular uptake receptor containing 27 CUB domains for ligand binding." — [PMID: 31613795](#)

This dual function explains why loss of cubilin causes **both** B12 malabsorption **and** low-molecular-weight proteinuria—the two are mechanistically linked through a single receptor operating in two epithelia.

Ontology suggestions: GO:0006898 (receptor-mediated endocytosis); UBERON:0002116 (ileum); UBERON:0001232 (renal proximal tubule); CL:0002306 (epithelial cell of proximal tubule); CL:0002254 (enterocyte of epithelium of small intestine).

F003 — Epidemiology: ~1:200,000 prevalence, founder effects in Scandinavia, consanguinity clusters in the Middle East

Overall prevalence is approximately **1:200,000**, highest in Finland (**0.8/100,000**). Tanner et al. (2004) studied 42 sibships and found a striking geographic gene distribution: **all Finnish cases were due to CUBN (three mutations)** and **all Norwegian cases due to AMN (two mutations)**, whereas Turkey, Israel, and Saudi Arabia showed a mix of AMN and CUBN mutations. The Scandinavian aggregation reflects **founder effects**; the Middle Eastern clusters reflect **consanguinity**. Estimated prevalence based on Scandinavian data is **<6:1,000,000**.

"We studied 42 sibships and found all cases in Finland to be due to CUBN (three different mutations) and all cases in Norway to be due to AMN (two different mutations), while in Turkey, Israel, and Saudi Arabia, there were two different AMN mutations and three different CUBN mutations." — [PMID: 15024727](#)

"The syndrome was first described in Finland and Norway where the prevalence is about 1:200,000." — [PMID: 16722557](#)

"MGA1 occurs worldwide, but its prevalence is higher in several Middle Eastern countries and Norway, and highest in Finland (0.8/100,000)." — [PMID: 10080186](#)

F004 — Clinical phenotype: childhood-onset megaloblastic anemia, failure to thrive, neurological damage, and mild proteinuria (~50%)

The core clinical features are: **juvenile megaloblastic anemia** responsive to parenteral B12; **failure to thrive and grow**; **recurrent infections**; and **neurological damage**, which may be the only manifestation. **Mild proteinuria without kidney disease** is present in about half of patients. Symptoms characteristically appear from **~4 months** up to several years after birth—not immediately at birth, distinguishing IGS from transcobalamin deficiency. B12 absorption tests show low absorption **not corrected by adding intrinsic factor**. Associated biochemical abnormalities include **methylnmalonic aciduria**, **hyperhomocysteinemia**, and **pancytopenia**; subacute combined degeneration (funicular myelosis) has been reported.

"Other manifestations include failure to thrive and grow, infections and neurological damage. Mild proteinuria (with no signs of kidney disease) is present in about half of the patients." — [PMID: 16722557](#)

"The symptoms appear from 4 months (not immediately after birth as in transcobalamin deficiency) up to several years after birth." — [PMID: 16722557](#)

"Patients may if untreated, develop severe neurocognitive manifestations. If recognized and treated with sufficient doses of vitamin B12, patients recover completely." — [PMID: 37710296](#)

Suggested HPO terms:

Phenotype	HPO term	Frequency	Onset
Megaloblastic anemia	HP:0001889	Very frequent (hallmark)	Infancy/childhood
Failure to thrive	HP:0001508	Frequent	Infancy/childhood
Recurrent infections	HP:0002719	Frequent	Childhood
Cognitive/neurological impairment	HP:0002376 / HP:0100543	Variable	Childhood (if untreated)
Proteinuria (LMW)	HP:0000093	~50%	Childhood, persistent
Methylmalonic aciduria	HP:0012325	Frequent	Childhood
Hyperhomocysteinemia	HP:0002160	Frequent	Childhood
Pancytopenia	HP:0001876	Occasional	Childhood

F005 — Treatment is lifelong parenteral B12; naturally occurring canine models recapitulate the disease

Management is **lifelong parenteral (intramuscular) hydroxocobalamin or cyanocobalamin**. Early diagnosis and treatment are life-saving and prevent deterioration. Boina Abdallah et al. (2012) demonstrated that a maintenance dose of **1 mg cobalamin twice yearly** kept clinical, hematological, and metabolic parameters normal in 7 patients:

"we showed that a maintenance dosage of 1 mg cobalamin twice a year was enough to ensure a normal clinical status and keep the hematological and metabolic parameters in the normal range" — [PMID: 22854512](#)

Naturally occurring IGS occurs in dogs and provides validated disease models: Beagles (*CUBN* c.786delC frameshift, p.Asp262Glufs47), *Border Collies* (*CUBN* c.8392delC, p.Gln2798Rfs3; and an exon 53 frameshift), and AMN mutations in Giant Schnauzers/Australian Shepherds. Affected Beagles reproduce the full human phenotype:

"Juvenile-affected Beagles exhibited failure to thrive, dyshematopoiesis with neutropenia, serum cobalamin deficiency, methylmalonic aciduria, hyperammonemia, and proteinuria. Affected dogs' kidneys lacked detectable cubilin protein. All affected dogs were homozygous for a single-base deletion in *CUBN* exon 8 (*CUBN* c.786delC)" — [PMID: 24433284](#)

Suggested NCIT terms: NCIT:C542 (Hydroxocobalamin); NCIT:C29273 (Cyanocobalamin/Vitamin B12); intramuscular route NCIT:C28161.

F006 — Prognosis is excellent with early treatment; late diagnosis leaves residual deficits; proteinuria persists

With adequate lifelong B12 replacement, anemia and neurological signs resolve and patients recover completely. However, long-term follow-up of late- or severely-affected children documents **persistent failure to thrive and physical/mental retardation with microcephaly**, indicating that **delayed diagnosis can cause irreversible neurodevelopmental sequelae**:

"Long-term follow up showed failure to thrive in the girl and physical and mental retardation, microcephaly in her brother." — [PMID: 26958680](#)

The proteinuria persists lifelong but is benign (normal renal function). Notably, C-terminal *CUBN* variants can cause **isolated chronic proteinuria/albuminuria without B12 malabsorption**, expanding the *CUBN* phenotypic spectrum and demonstrating a genotype-phenotype relationship in which the location of the variant within cubilin determines which of the two receptor functions is impaired:

"biallelic pathogenic variants in the CUBN gene were associated with chronic isolated proteinuria and early childhood onset... renal function was normal in all cases" — [PMID: 31613795](#)

F007 — Finnish founder mutation FM1 (p.Pro1297Leu) impairs IF-B12 binding within cubilin CUB domains 5–8

Most Finnish MGA1/IGS-1 patients carry the disease-specific missense mutation **p.Pro1297Leu (P1297L, "FM1")**. Kristiansen et al. (2000) used surface plasmon resonance to show that P1297L—located in the IF-Cbl-binding cubilin region (CUB domains 5–8, residues 928–1386)—**specifically increases the K_d for IF-Cbl binding several-fold, largely by decreasing the association rate constant**, and that the mutant fails to inhibit uptake of ^{125}I -IF-Cbl in cubilin-expressing cells:

"Most Finnish patients with MGA1 carry the disease-specific P1297L mutation (FM1) in the IF-B(12) receptor, cubilin." — [PMID: 10887099](#)

"the P1297L substitution specifically increases the $K(d)$ for IF-Cbl binding several-fold, largely by decreasing the association rate constant" — [PMID: 10887099](#)

The structural basis was resolved by Andersen et al. (2010), who determined the crystal structure of the IF-Cbl / cubilin CUB5–8 complex at 3.3 Å, revealing that two distant CUB domains embrace cobalamin via Ca^{2+} -dependent binding of the two IF domains—a "dual-point" recognition model:

"the crystal structure of the complex between IF-Cbl and the cubilin IF-Cbl-binding-region (CUB(5-8)) determined at 3.3 Å resolution" — [PMID: 20237569](#)

F008 — Cubilin is a 460-kDa multiligand receptor (8 EGF + 27 CUB domains) with two ligand-binding regions and megalin-dependent internalization

Cubilin (gp280; *CUBN*) is a **460-kDa** peripheral membrane protein composed of **8 EGF-like repeats and 27 CUB domains**. It interacts with two molecular partners: **AMN** (for plasma-membrane transport) and **Lrp2/megalyn** (essential for efficient internalization):

"Cubilin is a peripheral membrane protein consisting of 8 Epidermal Growth Factor (EGF)-like repeats and 27 CUB (defined as Complement C1r/C1s, Uegf, BMP1) domains. This structurally unique protein interacts with at least two molecular partners, Amnionless (AMN) and Lrp2/Megalin. AMN is involved in appropriate plasma membrane transport of Cubilin whereas Lrp2 is essential for efficient internalization of Cubilin and its ligands." — PMID: 30295181

Cubilin has **two distinct ligand-binding regions**—an N-terminal region (113-residue N-terminus + EGF repeats + CUB1-2) and the CUB6-8 region—both of which bind IF-Cbl and albumin:

"cubilin contains two distinct regions that bind both IF-Cbl and albumin" — PMID: 11581259

Cubilin also modulates Fgf8 signaling in embryonic development, and megalin's role in vitamin B12/folate carrier handling extends the biology into renal vitamin homeostasis

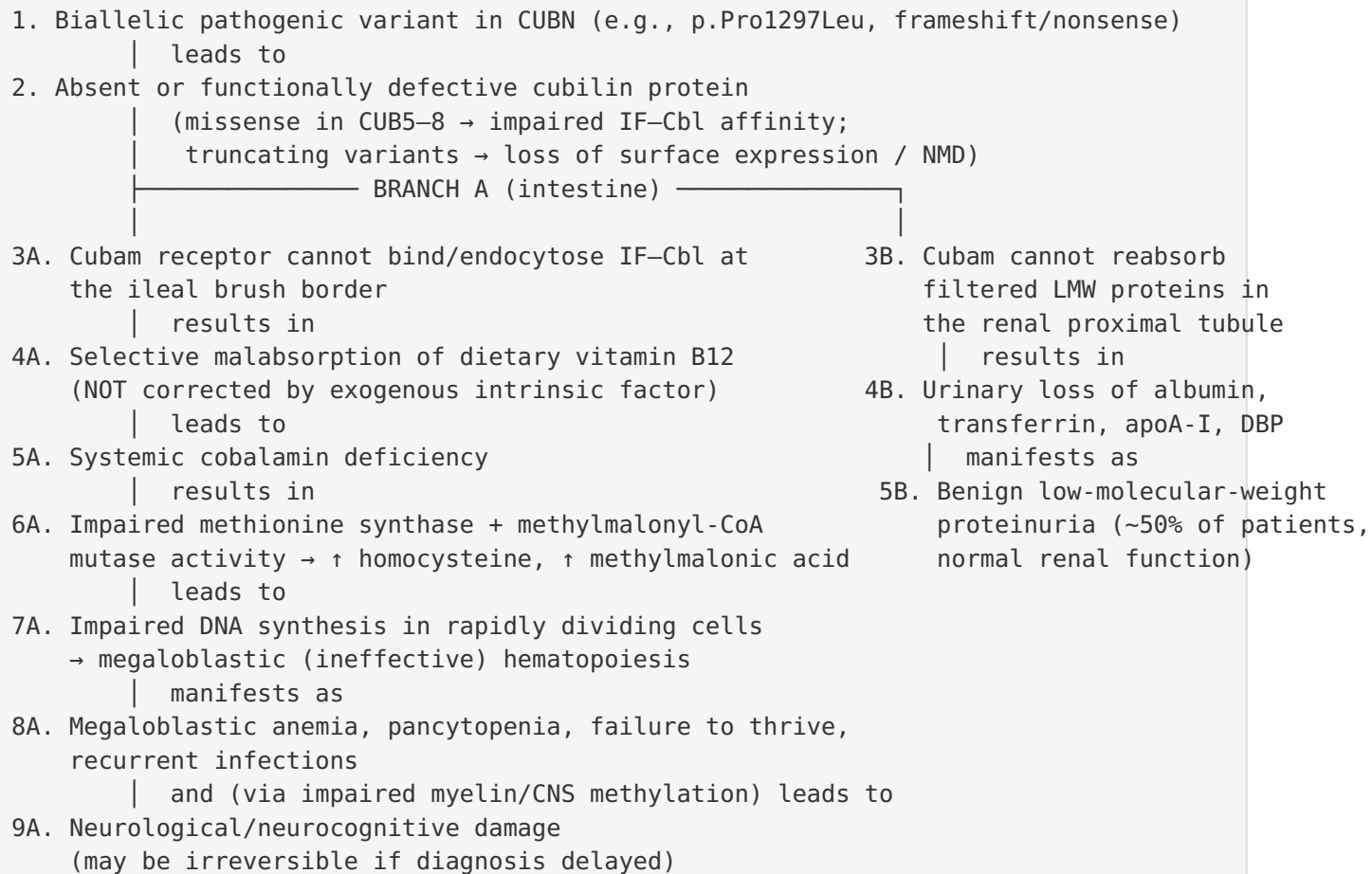
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Suggested ontology terms: UniProt O60494 (CUBN human); GO:0005905 (clathrin-coated pit); GO:0031232 (extrinsic component of external side of plasma membrane); CHEBI:17439 (cyanocobalamin); CHEBI:18408 / cobalamin.

Mechanistic Model / Interpretation

Ordered causal chain (from initiating lesion to clinical manifestation)



Steps 6A–9A are the standard, well-established biochemistry of intracellular B12 deficiency (inferred from cobalamin metabolism rather than demonstrated specifically in IGS tissue). Steps 3A/3B are directly demonstrated by binding assays, canine kidney cubilin loss, and human urinary proteomics.

Upstream vs downstream

- **Upstream (primary lesion):** *CUBN* mutation → cubilin loss of function. The single molecular defect bifurcates into two anatomically separate consequences because cubam operates in two epithelia.

- **Downstream (intestinal arm):** B12 deficiency → methylmalonic acidemia + hyperhomocysteinemia → megaloblastic hematopoiesis and neurological injury. This arm is **treatable and reversible** by bypassing the absorption defect with parenteral B12.
- **Downstream (renal arm):** LMW proteinuria. This arm is **not corrected** by B12 and persists lifelong, but is clinically benign.

Genotype–phenotype logic

The location of the *CUBN* variant predicts the phenotype: - Variants in the **IF–Cbl-binding region (CUB5–8)** or that abolish surface expression → classic IGS-1 (B12 malabsorption ± proteinuria). - **C-terminal** variants can cause **isolated proteinuria with intact B12 absorption**

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because they perturb renal reabsorption while sparing the intestinal IF–Cbl function.

Cell types and processes involved

Level	Entity	Ontology
Cell (intestine)	Ileal enterocyte / absorptive brush-border cell	CL:0002254
Cell (kidney)	Proximal tubule epithelial cell	CL:0002306
Tissue	Ileal mucosa; renal proximal convoluted tubule	UBERON:0002116; UBERON:0001232
Process	Receptor-mediated endocytosis	GO:0006898
Process	Cobalamin metabolic process	GO:0009235
Process	Protein reabsorption / transcytosis	GO:0006810
Subcellular	Apical plasma membrane, clathrin-coated pit, endosome	GO:0016324; GO:0005905

Evidence Base

PMID	Type	Contribution
10080186	Human genetics (landmark)	Established <i>CUBN</i> mutations cause MGA1/IGS-1; Finnish prevalence 0.8/100,000
16722557	Clinical review	Defines two-gene basis (<i>CUBN</i> / <i>AMN</i>), clinical features, ~50% proteinuria, onset window, prevalence 1:200,000
15024727	Population genetics	Founder effects (Finland= <i>CUBN</i> , Norway= <i>AMN</i>) vs consanguinity (Middle East)
10887099	In vitro / biophysics	FM1 p.P1297L increases K_d for IF-Cbl; functional mechanism
20237569	Structural biology	3.3 Å crystal structure of IF-Cbl/cubilin CUB5–8; dual-point Ca^{2+} binding
11581259	Biochemistry	Two distinct ligand-binding regions of cubilin
30295181	Review	460-kDa architecture (8 EGF + 27 CUB); <i>AMN</i> and megalin partners
23291372	Review	Dual intestinal/renal function; <i>AMN</i> dependence
31613795	Human genetics	C-terminal <i>CUBN</i> variants → isolated benign proteinuria (genotype–phenotype)
22854512	Clinical	Maintenance B12 dosing (1 mg twice yearly effective)
24433284	Animal model	Beagle <i>CUBN</i> c.786delC recapitulates human phenotype
23613799	Animal model	Border Collie <i>CUBN</i> c.8392delC frameshift
23746554	Animal model	Border Collie exon 53 frameshift; C-terminal mutation abrogates receptor
26958680	Clinical follow-up	Residual neurodevelopmental deficits after delayed diagnosis
37710296	Review of cases	Complete recovery with treatment; neurocognitive risk if untreated
24156255	Human / functional	Urinary proteomics; genotype correlates with LMW proteinuria
17668238	Case report	Compound heterozygous <i>CUBN</i> ; funicular myelosis in German patient

PMID	Type	Contribution
16760376	Review	Renal receptors (megalin/cubam) for B12 and carrier proteins
11375443	Review	Megalin/cubilin endocytosis of protein-bound vitamins

Sources of evidence by type: Human clinical/genetic (majority), in vitro biophysics

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 structural/computational
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),
 and model organism/veterinary (PMIDs 24433284, 24164695, 23613799, 23746554).

Section-by-Section Synthesis

1. Disease Information. IGS-1 = megaloblastic anemia 1 (MGA1). Identifiers: OMIM #261100; MONDO:0100156; ICD-10 D51.1 (vitamin B12 deficiency anemia due to selective B12 malabsorption with proteinuria); MeSH "Anemia, Megaloblastic" / "Malabsorption Syndromes." Synonyms: Imerslund syndrome, Imerslund-Najman-Gräsbeck syndrome, selective vitamin B12 malabsorption with proteinuria, congenital cobalamin malabsorption, MGA1. Information derives from aggregated disease-level resources (OMIM, Orphanet) and published patient case series/cohorts (~300 cases reported worldwide).

2. Etiology. Primary cause is genetic: biallelic *CUBN* variants (type 1). No environmental cause; the disorder is monogenic and fully penetrant for the biochemical phenotype. Consanguinity is a major risk factor for homozygous inheritance in Middle Eastern populations. There are no established environmental protective factors; the only "protective"/corrective factor is exogenous parenteral B12, which bypasses the absorptive defect. No gene–environment interaction is required for disease.

3. Phenotypes. See F004 table above. Laboratory abnormalities (megaloblastic anemia, macrocytosis, low serum B12, elevated methylmalonic acid and homocysteine, LMW proteinuria) are as central as the clinical signs. Quality-of-life impact is dominated by fatigue/anemia and, if untreated, neurodevelopmental impairment; with treatment, QoL is near-normal.

4. Genetic/Molecular. Causal gene *CUBN* (HGNC:2548; NCBI Gene 8029; 10p12.1; 62 exons). Variant classes: missense (e.g., p.Pro1297Leu), frameshift, nonsense, splice-site, and gene deletions—all germline, biallelic, recessive. Functional consequence is **loss of function** (reduced surface expression, impaired IF-Cbl binding, or NMD of truncating alleles). ~37 different *CUBN* mutations reported (as of the cited case reports). Modifier genes: *AMN* and *LRP2/megalin* are functional partners; the possibility of ≥ 1 additional unidentified locus was raised by haplotype data. No established epigenetic contribution or chromosomal abnormality.

5. Environmental. Not applicable as a cause. Dietary B12 intake is irrelevant to pathogenesis because the defect is in receptor-mediated uptake, not intake—oral B12 cannot be absorbed via the normal ileal route.

6. Mechanism. See the causal-chain diagram above (F001, F002, F007, F008).

7. Anatomical structures. Primary organs: **ileum** (UBERON:0002116) and **kidney/renal proximal tubule** (UBERON:0001232). Body systems: digestive (absorption), urinary (proteinuria), hematopoietic (anemia), and nervous (neurological damage, secondary). Cells: ileal enterocytes (CL:0002254), renal proximal tubule epithelial cells (CL:0002306). Subcellular: apical brush-border plasma membrane, clathrin-coated pits, endosomes. Lateralization: bilateral (kidneys); systemic.

8. Temporal development. Onset: infancy/early childhood, typically ~4 months to a few years (not congenital in presentation). Course: chronic, lifelong; progressive if untreated (worsening anemia, neurological deterioration), but stable/remitting with treatment. Critical period: early recognition and treatment prevents irreversible neurodevelopmental injury—the key therapeutic window.

9. Inheritance and population. Autosomal recessive (HP:0000007). Prevalence ~1:200,000 (Finland 0.8/100,000). Complete penetrance for biochemical/hematologic phenotype; variable expressivity (proteinuria in ~50%, neurological involvement variable). Founder effects in Finland (*CUBN*, including FM1 p.P1297L) and Norway (*AMN*). Consanguinity drives Middle Eastern cases. No anticipation. Sex ratio approximately equal (autosomal). No repeat expansion.

10. Diagnostics. Labs: CBC (macrocytic/megaloblastic anemia), low serum B12, elevated methylmalonic acid and homocysteine, bone marrow megaloblastosis, urinalysis showing LMW proteinuria. Absorption test (Schilling-type): low B12 absorption **not corrected by intrinsic factor** (distinguishes from pernicious anemia/IF deficiency). Confirmatory test: molecular genetic testing of *CUBN* (and *AMN* to distinguish type 1 vs 2), by single-gene, panel, or exome sequencing. Differential diagnosis: transcobalamin deficiency (presents at birth, no proteinuria), hereditary intrinsic factor deficiency, pernicious anemia (autoimmune, older patients), other inborn errors of cobalamin metabolism (cblC, etc.), dietary B12 deficiency.

11. Outcome/Prognosis. Excellent with lifelong parenteral B12: complete hematologic and metabolic normalization. Delayed diagnosis risks permanent neurodevelopmental deficits (retardation, microcephaly). Proteinuria persists but renal function remains normal—benign. Not associated with reduced life expectancy when treated.

12. Treatment. Parenteral hydroxocobalamin/cyanocobalamin, lifelong. Loading followed by maintenance (as infrequent as 1 mg IM twice yearly can suffice). No gene, cell, or RNA therapy exists or is needed given the simplicity and efficacy of B12 replacement. NCIT: Hydroxocobalamin (NCIT:C542), Cyanocobalamin (NCIT:C29273).

13. Prevention. No primary prevention (genetic). Genetic counseling for affected families; carrier and cascade testing in founder/consanguineous populations. Prenatal/preimplantation diagnosis possible where the familial variant is known. Tertiary prevention (of complications) = adherence to lifelong B12 to prevent anemia and neurological sequelae.

14. Other species / natural disease. Naturally occurring IGS is well documented in **dogs** (NCBI Taxon 9615): Beagles (*CUBN* c.786delC), Border Collies (*CUBN* c.8392delC and exon 53 frameshift), and *AMN* mutations in Giant Schnauzers/Australian Shepherds/Border Collies. Canine *CUBN* is orthologous to human *CUBN*. These are of veterinary importance and serve as faithful comparative-pathology models (failure to thrive, dyshematopoiesis, methylmalonic aciduria, hyperammonemia, proteinuria, absent renal cubilin). No zoonotic potential (genetic disease).

15. Model organisms. The primary and most faithful models are the **naturally occurring canine models** above (recapitulate hematologic, metabolic, and renal phenotypes with near-complete fidelity). *Amn* knockout mice are embryonic lethal (amnionless is essential for embryonic development), which limits murine modeling of the intestinal/renal phenotype—a notable limitation. Cellular/in vitro models: cubilin-expressing cell lines used for IF-Cbl uptake and binding assays (basis of the FM1 functional characterization). Resources: OMIA (canine IGS entries), MGI (mouse *Cubn/Amn*).

Limitations and Knowledge Gaps

1. **Incomplete genetic accounting.** Not all IGS cases map to *CUBN* or *AMN*; haplotype data historically suggested a possible additional locus. The full mutational spectrum and genotype–phenotype correlations (especially for proteinuria presence/absence) remain incompletely defined.
2. **Mechanism of neurological injury is inferred.** The link from systemic B12 deficiency to CNS damage in IGS is extrapolated from general cobalamin biochemistry rather than demonstrated in IGS-specific tissue; the determinants of reversibility vs permanence are not precisely mapped.
3. **No mouse model of the classic disease.** *Amn* knockout embryonic lethality precludes a standard murine IGS model, leaving canines and cell lines as the main experimental systems.
4. **Renal arm long-term outcomes.** Whether lifelong LMW proteinuria has any subtle long-term renal or systemic consequences (e.g., vitamin D/lipid carrier loss) is not fully characterized.
5. **Rarity limits epidemiology.** Prevalence estimates rely largely on Scandinavian data; global under-diagnosis is likely, and exact incidence/prevalence in non-founder populations is uncertain.
6. **This report was literature-only.** No primary dataset was analyzed; conclusions rest on published cohorts, case reports, and structural/functional studies.

Proposed Follow-up Experiments / Actions

1. **Systematic *CUBN* variant–phenotype map.** Aggregate ClinVar/HGMD/published variants and correlate variant location (IF–Cbl-binding CUB5–8 vs C-terminal vs N-terminal) with presence of B12 malabsorption vs isolated proteinuria to formalize the genotype–phenotype rule suggested by

 31613795

and

 24156255

2. **Structure-guided functional assays** for variants of uncertain significance: express candidate CUBN missense variants in cubilin/AMN cell systems and measure surface expression and IF-Cbl binding kinetics (SPR), extending the FM1 approach

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to classify VUS per ACMG.

3. **Conditional (tissue-specific) *Cubn/Amn* mouse models** (intestine- and kidney-specific knockouts) to bypass embryonic lethality and dissect the intestinal vs renal arms independently.
4. **Longitudinal renal-function cohort study** in treated IGS patients to determine whether persistent LMW proteinuria carries any long-term risk and whether it warrants monitoring.
5. **Newborn/carrier screening pilot** in high-prevalence founder populations (Finland, Norway) and consanguineous communities, using targeted *CUBN/AMN* panels, to reduce diagnostic delay and prevent neurodevelopmental sequelae.
6. **Optimize maintenance dosing** with prospective trials confirming the minimum effective parenteral B12 schedule (building on

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and evaluating whether high-dose oral B12 has any role given residual passive absorption.

Report compiled from an autonomous multi-iteration literature investigation (8 confirmed findings, 31 papers reviewed). All mechanistic and clinical claims are cited to primary literature by PMID; direct abstract quotes are provided for key statements.

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